

PROBLEM STATEMENT #29 | Smart Cities, Transport & Logistics

Municipal Infrastructure Damage Reporting App**1. Problem Context & Overview**

A citizen portal for reporting road potholes, broken streetlights, and water pipeline leaks using geotagged photos, with automated ticket dispatch to relevant municipal engineering departments.

Target Stakeholders / Actors: Citizen Reporter, Municipal Engineer

2. Sample Functional Requirement (FR) Guideline**Sample Requirement: FR-001 [Priority: High]**

Description: The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint incident location on the municipal GIS map.

Sample Acceptance Criteria: Pass: Pothole report routed to Road Infrastructure division with exact GPS coordinates; Fail: Report submitted without location data.

3. Sample Non-Functional Requirement (NFR) Guideline**Sample Requirement: NFR-001 [Type: Performance & Security]**

Description: The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]**• Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

• UML Use-Case Diagram:

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

• Use-Case Flow Specification:

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.

Use-Case Flow Specification

Actor(s): Citizen Reporter (primary), GIS/Mapping Service (supporting)

Trigger: Citizen taps "Report Damage" in the app

Preconditions:

- Citizen Reporter has the app installed and is logged in (or using guest mode).
- Device camera and, where available, location services are accessible to the app.

Postconditions:

- A new damage report ticket is created with a unique Ticket ID.
- The ticket is geotagged and routed to the correct municipal engineering department queue.
- The citizen receives on-screen confirmation and can track the ticket status.

Main Success Scenario:

1. Citizen Reporter selects "Report Damage" from the app home screen.
2. System prompts the citizen to capture or upload a photo of the damage.
3. Citizen captures/uploads a photo showing the pothole, broken streetlight, or pipeline leak.
4. System invokes UC-02 (Extract GPS Location) to read EXIF GPS metadata from the photo.
«include»
5. Citizen selects a damage category (Pothole / Streetlight / Water Leak) and adds an optional text description.
6. System validates that a photo, location, and category are present.
7. Citizen taps "Submit Report."
8. System creates a ticket with a unique Ticket ID, timestamps it, and pins it on the municipal GIS map.
9. System auto-dispatches the ticket to the relevant engineering department queue (UC-03).
10. System displays a confirmation screen with the Ticket ID to the citizen.
Use case ends successfully.

Alternate Flow — 4a. GPS Data Not Found in Photo

- System detects that the uploaded photo has no EXIF GPS metadata (e.g., location services were off).
- System prompts the citizen to manually drop a pin on the in-app map to mark the damage location.
- Citizen places the pin and confirms the location.
- Flow resumes at Main Success Scenario step 5.

Extension — Flag as Urgent (extends UC-01)

- If the citizen marks the damage as posing an immediate safety hazard, the optional "Flag as Urgent" extension inserts before step 8, tagging the ticket with High priority for expedited dispatch.